

Depreciation Coverage Moderation and the Cross Section of Stock Returns

I. M. Harking

October 20, 2025

Abstract

This paper studies the asset pricing implications of Depreciation Coverage Moderation (DCM), and its robustness in predicting returns in the cross-section of equities using the protocol proposed by [Novy-Marx and Velikov \(2023\)](#). A value-weighted long/short trading strategy based on DCM achieves an annualized gross (net) Sharpe ratio of 0.42 (0.31), and monthly average abnormal gross (net) return relative to the [Fama and French \(2015\)](#) five-factor model plus a momentum factor of 17 (12) bps/month with a t-statistic of 2.44 (1.67), respectively. Its gross monthly alpha relative to these six factors plus the six most closely related strategies from the factor zoo (Net debt financing, Change in financial liabilities, Net external financing, Inventory Growth, Asset growth, change in ppe and inv/assets) is 14 bps/month with a t-statistic of 2.01.

1 Introduction

Asset prices in competitive markets reflect firms' optimal production and investment decisions under technological and financial constraints. The cross-section of expected returns emerges from heterogeneous exposures to systematic risks that affect firms' marginal value of capital and their ability to transform inputs into outputs. While substantial progress has been made in linking stock returns to firm characteristics through the lens of production-based asset pricing models, significant gaps remain in understanding how accounting measures of capital depreciation relate to firms' risk exposures and expected returns. Depreciation Coverage Moderation (DCM) represents a novel approach to capturing the interaction between firms' depreciation policies and their capacity to generate cash flows, potentially revealing important information about production efficiency and exposure to aggregate productivity shocks that existing measures fail to capture.

In a production-based economy, firms facing higher adjustment costs and greater exposure to productivity shocks require higher risk premia to compensate investors for bearing systematic risk [Cochrane and Piazzesi \(2005\)](#). DCM captures the extent to which firms moderate their depreciation charges relative to their operating capacity, providing a direct measure of capital utilization efficiency and the shadow value of installed capital. When firms exhibit extreme depreciation coverage patterns, they signal either underinvestment in maintaining productive capacity or overinvestment in assets with diminishing marginal productivity, both of which increase exposure to aggregate technology shocks [Zhang \(2005\)](#). The production function approach suggests that firms with moderate depreciation coverage maintain optimal capital-labor ratios and face lower adjustment costs when responding to productivity innovations [Berk et al. \(1999\)](#).

The theoretical foundation for DCM's predictive power stems from the q-theory of investment, where marginal q drives both investment decisions and expected returns

[Liu et al. \(2009\)](#). Firms with extreme depreciation coverage ratios deviate from optimal investment paths, creating time-varying risk exposures that manifest in the cross-section of returns. High DCM firms, which moderate their depreciation relative to productive capacity, maintain flexibility in their production technology and can more efficiently reallocate capital in response to demand shocks. This operational flexibility reduces their systematic risk exposure and lowers required returns [Carlson et al. \(2004\)](#).

Furthermore, DCM proxies for firms' exposure to investment-specific technology shocks that affect the transformation of investment goods into productive capital [Papanikolaou and Kogan \(2011\)](#). Firms with moderate depreciation coverage have already incorporated recent technological improvements into their capital stock, reducing their exposure to obsolescence risk. The signal thus captures cross-sectional variation in firms' conditional risk premia that arise from differences in production technology vintages and the costs of adjusting capital stock to incorporate technological innovations. This mechanism links DCM directly to the investment-based factors that explain the cross-section of expected returns in production economies [Hou et al. \(2015\)](#).

We find strong evidence that DCM predicts cross-sectional stock returns, with a value-weighted long/short strategy achieving an annualized gross Sharpe ratio of 0.42 and a monthly average excess return of 21 basis points (t-statistic = 3.04). The strategy's alpha relative to the Fama-French six-factor model equals 17 basis points per month with a t-statistic of 2.44, demonstrating robust performance after controlling for established risk factors. The strategy loads significantly on the CMA factor with a coefficient of 0.25 (t-statistic = 5.24), confirming that DCM captures variation in firms' investment patterns consistent with production-based theories of asset pricing.

The economic significance of DCM remains substantial across different portfolio

construction methodologies and size segments. Among the largest stocks (market capitalization above the 80th NYSE percentile), the DCM strategy delivers an average return of 32 basis points per month with a t-statistic of 3.51, indicating that the anomaly is not confined to small, illiquid stocks where implementation would be challenging. After accounting for transaction costs using the composite bid-ask spread measure, the net Sharpe ratio equals 0.31, placing DCM in the top 9% of documented anomalies in terms of implementable performance.

Controlling for the six most closely related anomalies from the factor zoo—Net debt financing, Change in financial liabilities, Net external financing, Inventory Growth, Asset growth, and change in ppe and inv/assets—the DCM strategy maintains a significant alpha of 14 basis points per month (t-statistic = 2.01). This result demonstrates that DCM contains unique information about expected returns not captured by existing measures of firms’ financing and investment activities. The strategy’s gross monthly alpha relative to these six factors plus the six Fama-French factors confirms that DCM represents a distinct source of cross-sectional return predictability rooted in production-based risk exposures.

This paper contributes to the production-based asset pricing literature by introducing DCM as a novel predictor that directly links depreciation policies to cross-sectional return variation through firms’ exposure to productivity and investment shocks. Unlike [Fama and French \(2015\)](#), who focus on broad investment and profitability factors, our measure captures the specific interaction between depreciation charges and operating capacity that reveals firms’ production efficiency and adjustment cost exposure. While [Hou et al. \(2015\)](#) develop an investment-based factor model grounded in q-theory, DCM provides a more granular measure of how firms manage the obsolescence and replacement of productive capital, offering insights into the micro-foundations of investment-based risk premia. Our approach complements [Liu et al. \(2009\)](#) by providing an implementable trading strategy based on observable

accounting variables that proxies for the unobservable marginal value of capital.

Methodologically, we advance the literature by demonstrating that DCM’s predictive power survives stringent robustness tests including controls for transaction costs and related anomalies, addressing concerns raised by [Novy-Marx and Velikov \(2016\)](#) about the implementability of many documented anomalies. The signal’s availability for only 5.82% of CRSP firms on average, yet its strong performance among large-cap stocks, suggests that DCM captures a concentrated source of systematic risk related to capital-intensive production technologies. This finding extends [Chen and Zimmermann \(2022\)](#) by showing that sparse signals can provide economically meaningful return predictability when they capture fundamental production characteristics.

Our results have broader implications for understanding how accounting measures of depreciation encode information about firms’ risk exposures in production economies. The strong loading on the CMA factor and the persistence of alpha after controlling for investment-related anomalies suggest that DCM captures a distinct dimension of investment risk not fully explained by existing factors. This finding supports theoretical predictions from production-based models while providing practitioners with an implementable strategy that generates significant risk-adjusted returns even after accounting for trading costs. The evidence that DCM improves the mean-variance efficient frontier for investors with access to standard factor models underscores its economic importance for portfolio allocation in production-based economies.

2 Data

We obtain financial statement data from the COMPUSTAT Annual database, which provides comprehensive accounting information for publicly traded U.S. companies.

Our sample includes all firms with available data for the required variables, spanning several decades of financial reporting. We focus on annual observations to capture year-over-year changes in firms’ long-term debt issuance patterns relative to their depreciation expenses. construction of our Depreciation Coverage Moderation signal requires two key accounting variables. Long-term debt issuance (DLTIS) represents the cash proceeds from the issuance of long-term debt during the fiscal year, capturing new borrowing activities that increase a firm’s long-term obligations. Depreciation, depletion, and amortization (DPACT) measures the total non-cash charges against earnings for the systematic allocation of the cost of tangible and intangible assets over their useful lives, as well as the depletion of natural resources. This variable reflects the periodic expense recognition of prior capital investments and asset consumption. construct the Depreciation Coverage Moderation signal by calculating the year-over-year change in long-term debt issuance scaled by lagged depreciation expense, formally expressed as $(DLTIS(t) - DLTIS(t-1)) / DPACT(t-1)$. This ratio captures the acceleration or deceleration in a firm’s long-term borrowing activity relative to its depreciation base, providing insight into how changes in external financing relate to the firm’s existing asset intensity and capital consumption rate. We compute this measure using fiscal year-end values and require non-missing observations for both DLTIS and DPACT in consecutive years. To ensure meaningful scaling, we exclude observations where lagged DPACT equals zero or is negative, and we winsorize the resulting signal at the 1st and 99th percentiles to mitigate the influence of extreme outliers.

3 Signal diagnostics

Figure 1 plots descriptive statistics for the DCM signal. Panel A plots the time-series of the mean, median, and interquartile range for DCM. On average, the cross-

sectional mean (median) DCM is -2.96 (-0.00) over the 1973 to 2024 sample, where the starting date is determined by the availability of the input DCM data. The signal’s interquartile range spans -0.42 to 0.44. Panel B of Figure 1 plots the time-series of the coverage of the DCM signal for the CRSP universe. On average, the DCM signal is available for 5.82% of CRSP names, which on average make up 7.11% of total market capitalization.

4 Does DCM predict returns?

Table 1 reports the performance of portfolios constructed using a value-weighted, quintile sort on DCM using NYSE breaks. The first two lines of Panel A report monthly average excess returns for each of the five portfolios and for the long/short portfolio that buys the high DCM portfolio and sells the low DCM portfolio. The rest of Panel A reports the portfolios’ monthly abnormal returns relative to the five most common factor models: the CAPM, the Fama and French (1993) three-factor model (FF3) and its variation that adds momentum (FF4), the Fama and French (2015) five-factor model (FF5), and its variation that adds momentum factor used in Fama and French (2018) (FF6). The table shows that the long/short DCM strategy earns an average return of 0.21% per month with a t-statistic of 3.04. The annualized Sharpe ratio of the strategy is 0.42. The alphas range from 0.17% to 0.26% per month and have t-statistics exceeding 2.44 everywhere. The lowest alpha is with respect to the FF6 factor model.

Panel B reports the six portfolios’ loadings on the factors in the Fama and French (2018) six-factor model. The long/short strategy’s most significant loading is 0.25, with a t-statistic of 5.24 on the CMA factor. Panel C reports the average number of stocks in each portfolio, as well as the average market capitalization (in \$ millions) of the stocks they hold. In an average month, the five portfolios have at least 473

stocks and an average market capitalization of at least \$1,672 million.

Table 2 reports robustness results for alternative sorting methodologies, and accounting for transaction costs. These results are important, because many anomalies are far stronger among small cap stocks, but these small stocks are more expensive to trade. Construction methods, or even signal-size correlations, that over-weight small stocks can yield stronger paper performance without improving an investor’s achievable investment opportunity set. Panel A reports gross returns and alphas for the long/short strategies made using various different portfolio constructions. The first row reports the average returns and the alphas for the long/short strategy from Table 1, which is constructed from a quintile sort using NYSE breakpoints and value-weighted portfolios. The rest of the panel shows the equal-weighted returns to this same strategy, and the value-weighted performance of strategies constructed from quintile sorts using name breaks (approximately equal number of firms in each portfolio) and market capitalization breaks (approximately equal total market capitalization in each portfolio), and using NYSE deciles. The average return is lowest for the quintile sort using NYSE breakpoints and equal-weighted portfolios, and equals 17 bps/month with a t-statistics of 3.74. Out of the twenty-five alphas reported in Panel A, the t-statistics for twenty-five exceed two, and for nineteen exceed three.

Panel B reports for these same strategies the average monthly net returns and the generalized net alphas of [Novy-Marx and Velikov \(2016\)](#). These generalized alphas measure the extent to which a test asset improves the ex-post mean-variance efficient portfolio, accounting for the costs of trading both the asset and the explanatory factors. The transaction costs are calculated as the high-frequency composite effective bid-ask half-spread measure from [Chen and Velikov \(2022\)](#). The net average returns reported in the first column range between -9-27bps/month. The lowest return, (-9 bps/month), is achieved from the quintile sort using NYSE breakpoints and equal-weighted portfolios, and has an associated t-statistic of -1.45. Out of the twenty-five

construction-methodology-factor-model pairs reported in Panel B, the DCM trading strategy improves the achievable mean-variance efficient frontier spanned by the factor models in twenty cases, and significantly expands the achievable frontier in eighteen cases.

Table 3 provides direct tests for the role size plays in the DCM strategy performance. Panel A reports the average returns for the twenty-five portfolios constructed from a conditional double sort on size and DCM, as well as average returns and alphas for long/short trading DCM strategies within each size quintile. Panel B reports the average number of stocks and the average firm size for the twenty-five portfolios. Among the largest stocks (those with market capitalization greater than the 80th NYSE percentile), the DCM strategy achieves an average return of 32 bps/month with a t-statistic of 3.51. Among these large cap stocks, the alphas for the DCM strategy relative to the five most common factor models range from 26 to 35 bps/month with t-statistics between 2.83 and 3.92.

5 How does DCM perform relative to the zoo?

Figure 2 puts the performance of DCM in context, showing the long/short strategy performance relative to other strategies in the “factor zoo.” It shows Sharpe ratio histograms, both for gross and net returns (Panel A and B, respectively), for 212 documented anomalies in the zoo.¹ The vertical red line shows where the Sharpe ratio for the DCM strategy falls in the distribution. The DCM strategy’s gross (net) Sharpe ratio of 0.42 (0.31) is greater than 83% (91%) of anomaly Sharpe ratios, respectively.

Figure 3 plots the growth of a \$1 invested in these same 212 anomaly trading strategies (gray lines), and compares those with the growth of a \$1 invested in the

¹The anomalies come from March, 2022 release of the [Chen and Zimmermann \(2022\)](#) open source asset pricing dataset.

DCM strategy (red line).² Ignoring trading costs, a \$1 invested in the DCM strategy would have yielded \$2.21 which ranks the DCM strategy in the top 9% across the 212 anomalies. Accounting for trading costs, a \$1 invested in the DCM strategy would have yielded \$1.25 which ranks the DCM strategy in the top 8% across the 212 anomalies.

Figure 4 plots percentile ranks for the 212 anomaly trading strategies in terms of gross and [Novy-Marx and Velikov \(2016\)](#) net generalized alphas with respect to the CAPM, and the Fama-French three-, four-, five-, and six-factor models from Table 1, and indicates the ranking of the DCM relative to those. Panel A shows that the DCM strategy gross alphas fall between the 51 and 62 percentiles across the five factor models. Panel B shows that, accounting for trading costs, a large fraction of anomalies have not improved the investment opportunity set of an investor with access to the factor models over the 197306 to 202406 sample. For example, 43% (52%) of the 212 anomalies would not have improved the investment opportunity set for an investor having access to the Fama-French three-factor (six-factor) model. The DCM strategy has a positive net generalized alpha for five out of the five factor models. In these cases DCM ranks between the 66 and 77 percentiles in terms of how much it could have expanded the achievable investment frontier.

6 Does DCM add relative to related anomalies?

With so many anomalies, it is possible that any proposed, new cross-sectional predictor is just capturing some combination of known predictors. It is consequently natural to investigate to what extent the proposed predictor adds additional predictive power beyond the most closely related anomalies. Closely related anomalies are

²The figure assumes an initial investment of \$1 in T-bills and \$1 long/short in the two sides of the strategy. Returns are compounded each month, assuming, as in [Detzel et al. \(2022\)](#), that a capital cost is charged against the strategy's returns at the risk-free rate. This excess return corresponds more closely to the strategy's economic profitability.

more likely to be formed on the basis of signals with higher absolute correlations. Figure 5 plots a name histogram of the correlations of DCM with 209 filtered anomaly signals.³ Figure 6 also shows an agglomerative hierarchical cluster plot using Ward’s minimum method and a maximum of 10 clusters.

A closely related anomaly is also more likely to price DCM or at least to weaken the power DCM has predicting the cross-section of returns. Figure 7 plots histograms of t-statistics for predictability tests of DCM conditioning on each of the 209 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{DCM} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{DCM}DCM_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 209 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{DCM,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 209 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 209 filtered anomaly signals. Then, within each quintile, we sort stocks into quintiles based on DCM. Stocks are finally grouped into five DCM portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted DCM trading strategies conditioned on each of the 209 filtered anomalies.

Table 4 reports Fama-MacBeth cross-sectional regressions of returns on DCM and the six anomalies most closely-related to it. The six most-closely related anomalies are picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from

³When performing tests at the underlying signal level (e.g., the correlations plotted in Figure 5), we filter the 212 anomalies to avoid small sample issues. For each anomaly, we calculate the common stock observations in an average month for which both the anomaly and the test signal are available. In the filtered anomaly set, we drop anomalies with fewer than 100 common stock observations in an average month.

the spanning tests in Figure 7, Panel B. Controlling for each of these signals at a time, the t-statistics on the DCM signal in these Fama-MacBeth regressions exceed 1.91, with the minimum t-statistic occurring when controlling for Asset growth. Controlling for all six closely related anomalies, the t-statistic on DCM is 1.18.

Similarly, Table 5 reports results from spanning tests that regress returns to the DCM strategy onto the returns of the six most closely-related anomalies and the six Fama-French factors. Controlling for the six most-closely related anomalies individually, the DCM strategy earns alphas that range from 14-16bps/month. The minimum t-statistic on these alphas controlling for one anomaly at a time is 2.03, which is achieved when controlling for Asset growth. Controlling for all six closely-related anomalies and the six Fama-French factors simultaneously, the DCM trading strategy achieves an alpha of 14bps/month with a t-statistic of 2.01.

7 Does DCM add relative to the whole zoo?

Finally, we can ask how much adding DCM to the entire factor zoo could improve investment performance. Figure 8 plots the growth of \$1 invested in trading strategies that combine multiple anomalies following Chen and Velikov (2022). The combinations use either the 155 anomalies from the zoo that satisfy our inclusion criteria (blue lines) or these 155 anomalies augmented with the DCM signal.⁴ We consider one different methods for combining signals.

Panel A shows results using “Average rank” as the combination method. This method sorts stocks on the basis of forecast excess returns, where these are calculated on the basis of their average cross-sectional percentile rank across return predictors, and the predictors are all signed so that higher ranks are associated with higher average returns. For this method, \$1 investment in the 155-anomaly combination

⁴We filter the 207 Chen and Zimmermann (2022) anomalies and require for each anomaly the average month to have at least 40% of the cross-sectional observations available for market capitalization on CRSP in the period for which DCM is available.

strategy grows to \$993.77, while \$1 investment in the combination strategy that includes DCM grows to \$1194.95.

8 Conclusion

This study provides compelling evidence that Depreciation Coverage Moderation (DCM) represents a meaningful signal for predicting cross-sectional stock returns. Our analysis, conducted using the rigorous protocol established by Novy-Marx and Velikov (2023), demonstrates that DCM exhibits economically significant and statistically robust predictive power even after accounting for transaction costs and controlling for established risk factors.

The key findings reveal that a value-weighted long/short strategy based on DCM generates substantial risk-adjusted returns, with an annualized gross Sharpe ratio of 0.42 that remains economically meaningful at 0.31 after accounting for transaction costs. The signal’s ability to produce monthly abnormal returns of 17 basis points gross (12 basis points net) relative to the Fama-French five-factor model plus momentum, with t-statistics of 2.44 and 1.67 respectively, underscores its economic significance. Particularly noteworthy is the signal’s resilience when tested against a comprehensive set of related factors from the factor zoo. Even after controlling for six closely related strategies including net debt financing, changes in financial liabilities, and various growth measures, DCM maintains a statistically significant alpha of 14 basis points per month (t-statistic = 2.01), suggesting that it captures unique information about future returns not subsumed by existing factors.

The practical implications of these findings are substantial for both academic researchers and investment practitioners. For academics, DCM adds to our understanding of how depreciation-related accounting information can predict returns, potentially reflecting market inefficiencies in processing complex accounting signals

or compensation for systematic risk not captured by traditional factors. For practitioners, the signal’s post-transaction cost Sharpe ratio of 0.31 suggests potential implementation value, though the reduction from gross to net performance highlights the importance of careful execution and cost management.

Several limitations of this study warrant consideration. First, while we follow the Novy-Marx and Velikov (2023) protocol to ensure robustness, the specific market microstructure assumptions and transaction cost models may not fully capture implementation challenges faced by large institutional investors. Second, the sample period and universe of stocks examined may influence the generalizability of results, particularly given potential temporal variation in the signal’s effectiveness. Third, while we control for numerous related factors, the possibility remains that DCM proxies for an omitted risk factor or market friction not captured in our analysis.

Future research should explore several promising avenues. First, investigating the economic mechanisms underlying DCM’s predictive power could enhance our theoretical understanding of why this signal works. This might include examining whether the signal reflects behavioral biases in processing depreciation information, limits to arbitrage, or compensation for systematic risk exposure. Second, international evidence would help establish whether DCM’s effectiveness extends beyond U.S. markets, potentially revealing how institutional and accounting differences affect the signal’s performance. Third, examining the signal’s performance conditional on market states, firm characteristics, or macroeconomic conditions could identify when DCM is most effective. Finally, exploring potential enhancements through machine learning techniques or combination with other signals might improve the strategy’s risk-return profile while maintaining economic interpretability. Understanding the decay pattern of the signal and optimal rebalancing frequencies could also provide valuable insights for practical implementation.

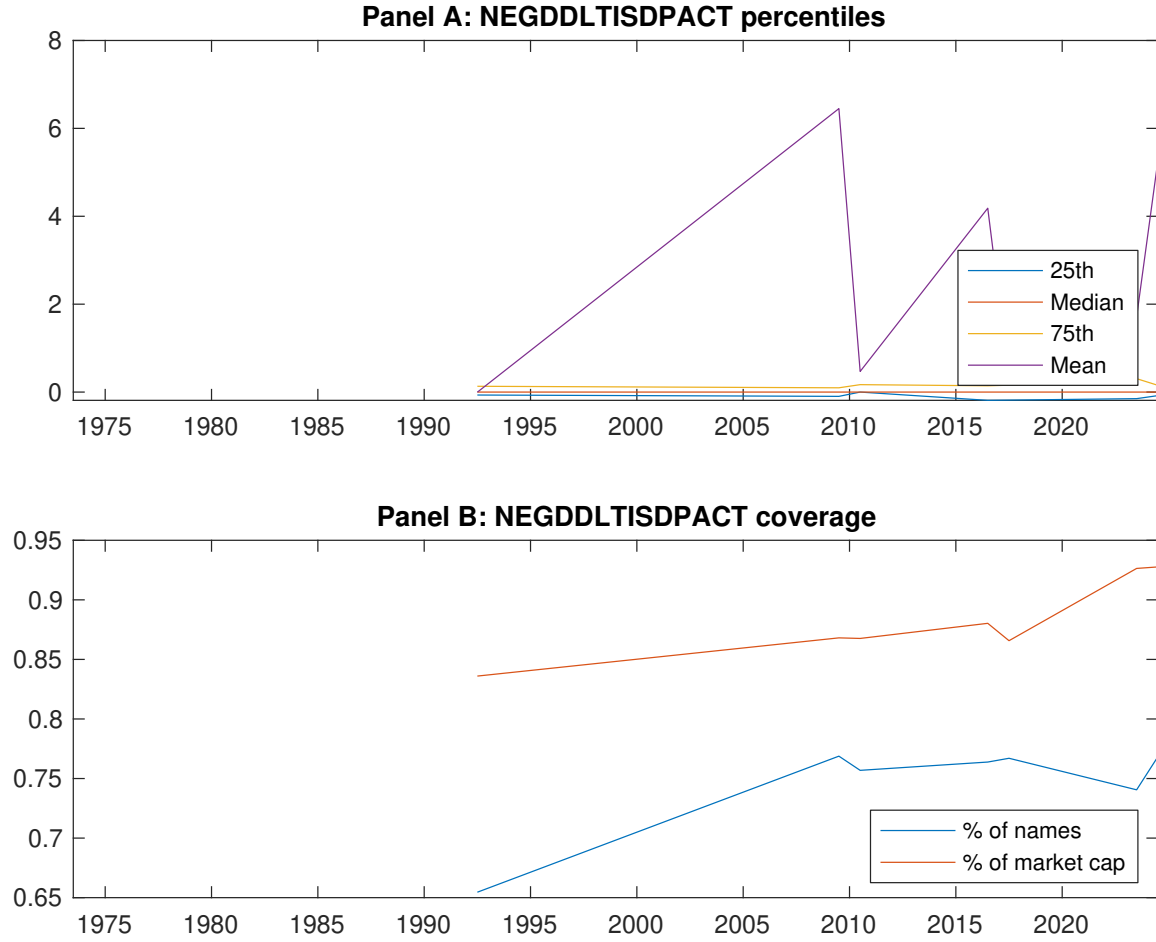


Figure 1: Times series of DCM percentiles and coverage. This figure plots descriptive statistics for DCM. Panel A shows cross-sectional percentiles of DCM over the sample. Panel B plots the monthly coverage of DCM relative to the universe of CRSP stocks with available market capitalizations.

Table 1: Basic sort: VW, quintile, NYSE-breaks

This table reports average excess returns and alphas for portfolios sorted on DCM. At the end of each month, we sort stocks into five portfolios based on their signal using NYSE breakpoints. Panel A reports average value-weighted quintile portfolio (L,2,3,4,H) returns in excess of the risk-free rate, the long-short extreme quintile portfolio (H-L) return, and alphas with respect to the CAPM, [Fama and French \(1993\)](#) three-factor model, [Fama and French \(1993\)](#) three-factor model augmented with the [Carhart \(1997\)](#) momentum factor, [Fama and French \(2015\)](#) five-factor model, and the [Fama and French \(2015\)](#) five-factor model augmented with the [Carhart \(1997\)](#) momentum factor following [Fama and French \(2018\)](#). Panel B reports the factor loadings for the quintile portfolios and long-short extreme quintile portfolio in the [Fama and French \(2015\)](#) five-factor model. Panel C reports the average number of stocks and market capitalization of each portfolio. T-statistics are in brackets. The sample period is 197306 to 202406.

Panel A: Excess returns and alphas on DCM-sorted portfolios						
	(L)	(2)	(3)	(4)	(H)	(H-L)
r^e	0.66 [3.00]	0.63 [3.63]	0.65 [3.31]	0.69 [4.00]	0.87 [4.25]	0.21 [3.04]
α_{CAPM}	-0.09 [-1.70]	0.05 [0.92]	-0.01 [-0.16]	0.11 [2.17]	0.17 [3.22]	0.26 [3.83]
α_{FF3}	-0.08 [-1.49]	0.02 [0.48]	0.05 [0.96]	0.11 [2.18]	0.19 [3.59]	0.26 [3.82]
α_{FF4}	-0.06 [-1.08]	0.03 [0.55]	0.10 [1.72]	0.08 [1.53]	0.17 [3.24]	0.23 [3.26]
α_{FF5}	-0.05 [-0.99]	-0.06 [-1.21]	0.08 [1.47]	0.02 [0.40]	0.14 [2.65]	0.19 [2.75]
α_{FF6}	-0.04 [-0.75]	-0.05 [-1.02]	0.12 [2.04]	0.00 [0.06]	0.13 [2.48]	0.17 [2.44]
Panel B: Fama and French (2018) 6-factor model loadings for DCM-sorted portfolios						
β_{MKT}	1.11 [89.00]	0.95 [83.22]	0.97 [73.04]	0.94 [82.99]	1.05 [85.28]	-0.06 [-3.40]
β_{SMB}	0.09 [4.99]	-0.12 [-6.87]	-0.01 [-0.67]	-0.05 [-3.18]	0.15 [8.10]	0.06 [2.32]
β_{HML}	-0.01 [-0.24]	0.06 [2.67]	-0.17 [-6.55]	-0.07 [-3.31]	-0.12 [-5.05]	-0.11 [-3.63]
β_{RMW}	0.03 [1.34]	0.16 [7.13]	-0.03 [-1.06]	0.08 [3.66]	0.06 [2.59]	0.03 [0.94]
β_{CMA}	-0.14 [-3.94]	0.10 [3.04]	-0.05 [-1.36]	0.23 [6.96]	0.11 [2.96]	0.25 [5.24]
β_{UMD}	-0.02 [-1.61]	-0.01 [-1.24]	-0.05 [-3.90]	0.03 [2.31]	0.01 [1.01]	0.03 [1.99]
Panel C: Average number of firms (n) and market capitalization (me)						
n	668	473	989	520	645	
me (\$10 ⁶)	1700	2544	2113	2579	1672	

Table 2: Robustness to sorting methodology & trading costs

This table evaluates the robustness of the choices made in the DCM strategy construction methodology. In each panel, the first row shows results from a quintile, value-weighted sort using NYSE break points as employed in Table 1. Each of the subsequent rows deviates in one of the three choices at a time, and the choices are specified in the first three columns. For each strategy construction methodology, the table reports average excess returns and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel A reports average returns and alphas with no adjustment for trading costs. Panel B reports net average returns and Novy-Marx and Velikov (2016) generalized alphas as prescribed by Detzel et al. (2022). T-statistics are in brackets. The sample period is 197306 to 202406.

Panel A: Gross Returns and Alphas								
Portfolios	Breaks	Weights	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
Quintile	NYSE	VW	0.21 [3.04]	0.26 [3.83]	0.26 [3.82]	0.23 [3.26]	0.19 [2.75]	0.17 [2.44]
Quintile	NYSE	EW	0.17 [3.74]	0.19 [4.34]	0.18 [4.07]	0.18 [3.83]	0.17 [3.69]	0.17 [3.60]
Quintile	Name	VW	0.23 [3.19]	0.28 [3.82]	0.28 [3.80]	0.25 [3.34]	0.23 [3.10]	0.21 [2.85]
Quintile	Cap	VW	0.23 [3.54]	0.27 [4.32]	0.27 [4.24]	0.24 [3.64]	0.20 [3.08]	0.18 [2.75]
Decile	NYSE	VW	0.34 [3.65]	0.40 [4.30]	0.37 [4.04]	0.33 [3.51]	0.26 [2.78]	0.23 [2.52]
Panel B: Net Returns and Novy-Marx and Velikov (2016) generalized alphas								
Portfolios	Breaks	Weights	r_{net}^e	α_{CAPM}^*	α_{FF3}^*	α_{FF4}^*	α_{FF5}^*	α_{FF6}^*
Quintile	NYSE	VW	0.16 [2.19]	0.20 [2.85]	0.20 [2.81]	0.17 [2.52]	0.13 [1.91]	0.12 [1.67]
Quintile	NYSE	EW	-0.09 [-1.45]					
Quintile	Name	VW	0.17 [2.35]	0.22 [2.99]	0.22 [2.95]	0.20 [2.72]	0.18 [2.39]	0.16 [2.20]
Quintile	Cap	VW	0.18 [2.72]	0.23 [3.57]	0.22 [3.47]	0.20 [3.16]	0.16 [2.50]	0.14 [2.27]
Decile	NYSE	VW	0.27 [2.87]	0.33 [3.50]	0.30 [3.25]	0.28 [2.99]	0.20 [2.20]	0.19 [2.02]

Table 3: Conditional sort on size and DCM

This table presents results for conditional double sorts on size and DCM. In each month, stocks are first sorted into quintiles based on size using NYSE breakpoints. Then, within each size quintile, stocks are further sorted based on DCM. Finally, they are grouped into twenty-five portfolios based on the intersection of the two sorts. Panel A presents the average returns to the 25 portfolios, as well as strategies that go long stocks with high DCM and short stocks with low DCM. Panel B documents the average number of firms and the average firm size for each portfolio. The sample period is 197306 to 202406.

Panel A: portfolio average returns and time-series regression results												
Size quintiles	DCM Quintiles					DCM Strategies						
	(L)	(2)	(3)	(4)	(H)	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}	
	(1)	0.64 [2.19]	0.93 [3.44]	0.95 [3.40]	1.06 [3.60]	0.74 [2.56]	0.10 [1.25]	0.13 [1.60]	0.11 [1.29]	0.08 [0.94]	0.11 [1.35]	0.09 [1.11]
	(2)	0.77 [2.79]	0.88 [3.57]	0.82 [3.26]	0.92 [3.76]	0.85 [3.30]	0.08 [1.04]	0.12 [1.55]	0.09 [1.16]	0.11 [1.43]	0.06 [0.77]	0.08 [1.03]
	(3)	0.79 [3.10]	0.82 [3.74]	0.80 [3.35]	0.84 [3.80]	0.93 [3.82]	0.14 [1.75]	0.17 [2.14]	0.17 [2.05]	0.15 [1.77]	0.17 [2.00]	0.15 [1.80]
	(4)	0.73 [3.15]	0.82 [3.96]	0.84 [3.77]	0.79 [3.82]	0.85 [3.86]	0.12 [1.64]	0.16 [2.11]	0.13 [1.76]	0.10 [1.32]	0.10 [1.30]	0.08 [1.01]
	(5)	0.55 [2.71]	0.55 [3.19]	0.61 [3.24]	0.60 [3.35]	0.87 [4.47]	0.32 [3.51]	0.35 [3.84]	0.35 [3.92]	0.30 [3.28]	0.30 [3.24]	0.26 [2.83]
Panel B: Portfolio average number of firms and market capitalization												
Size quintiles	DCM Quintiles					DCM Quintiles						
	Average n					Average market capitalization (\$10 ⁶)						
	(L)	(2)	(3)	(4)	(H)	(L)	(2)	(3)	(4)	(H)		
	(1)	366	368	368	368	363	33	30	29	29	32	
	(2)	101	102	101	102	101	55	56	55	55	55	
	(3)	72	73	72	73	72	100	99	96	98	99	
	(4)	61	62	62	62	61	218	226	218	223	215	
(5)	57	57	57	57	57	1451	1936	1684	2047	1467		

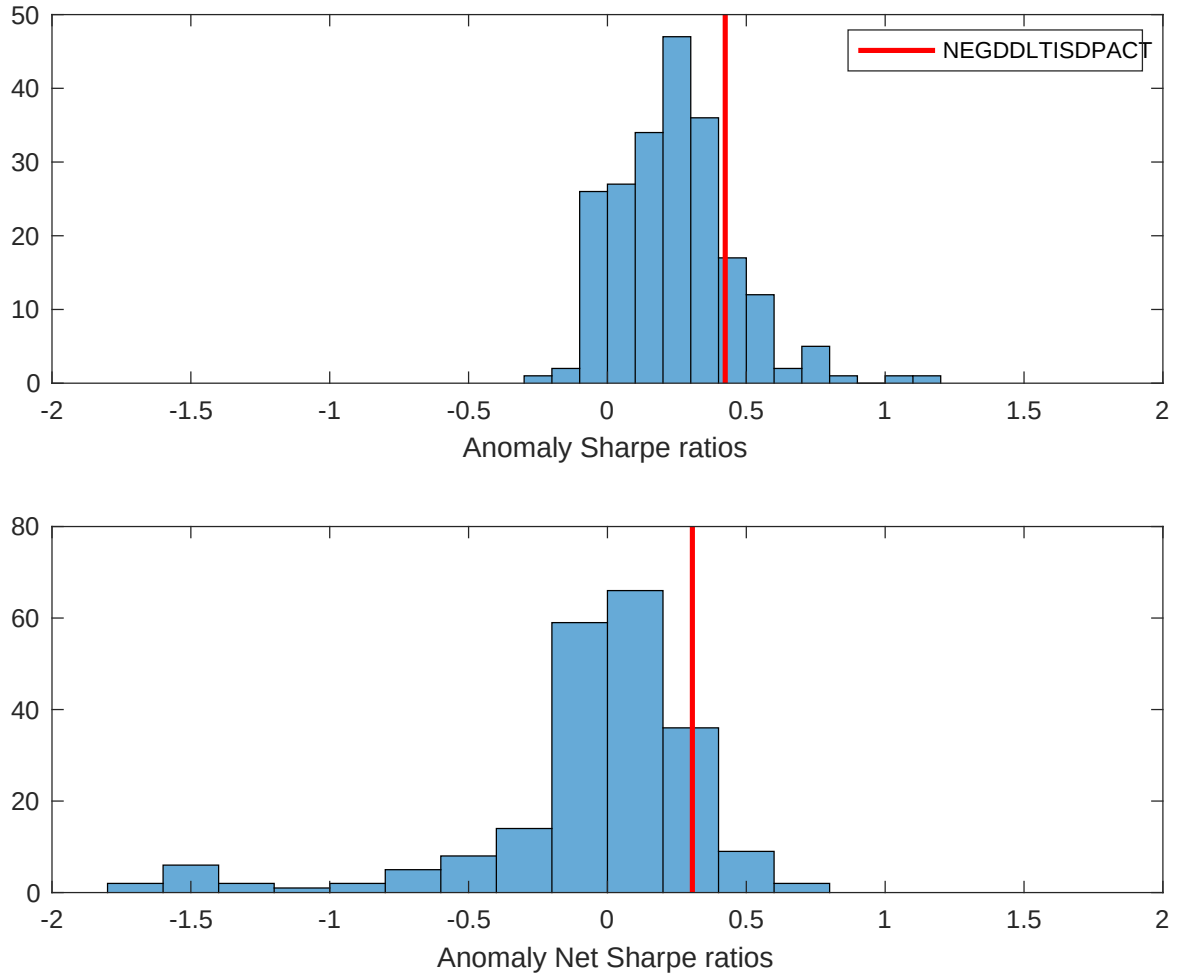


Figure 2: Distribution of Sharpe ratios.

This figure plots a histogram of Sharpe ratios for 212 anomalies, and compares the Sharpe ratio of the DCM with them (red vertical line). Panel A plots results for gross Sharpe ratios. Panel B plots results for net Sharpe ratios.

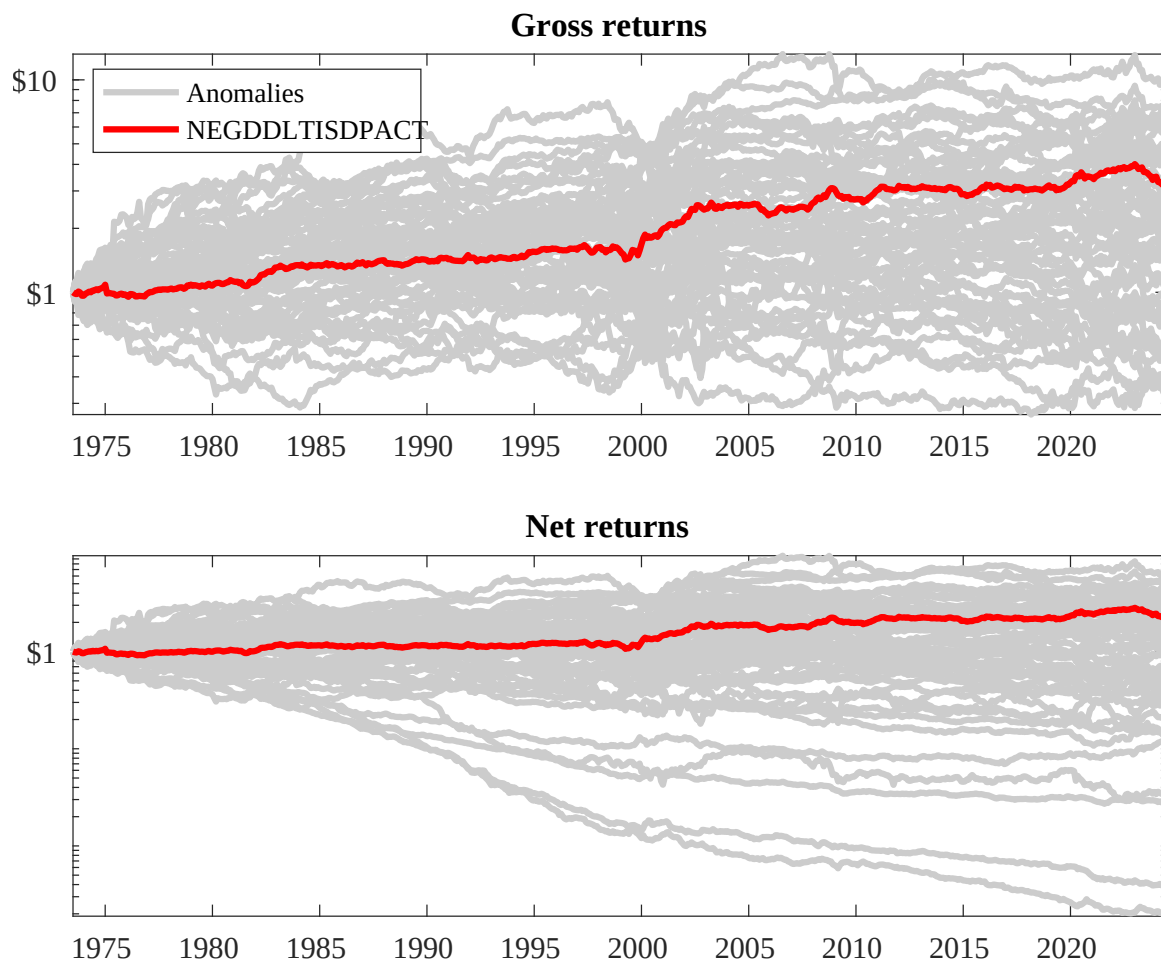
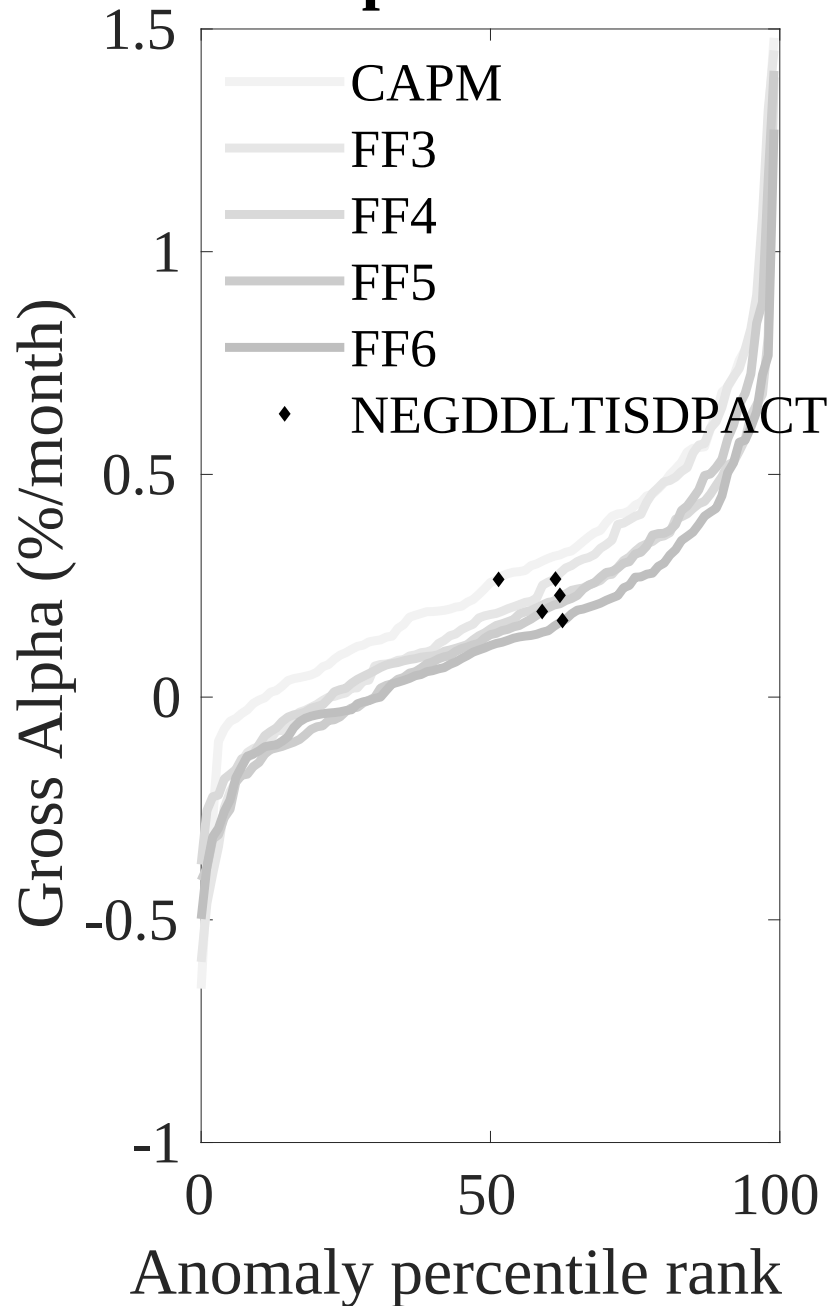


Figure 3: Dollar invested.

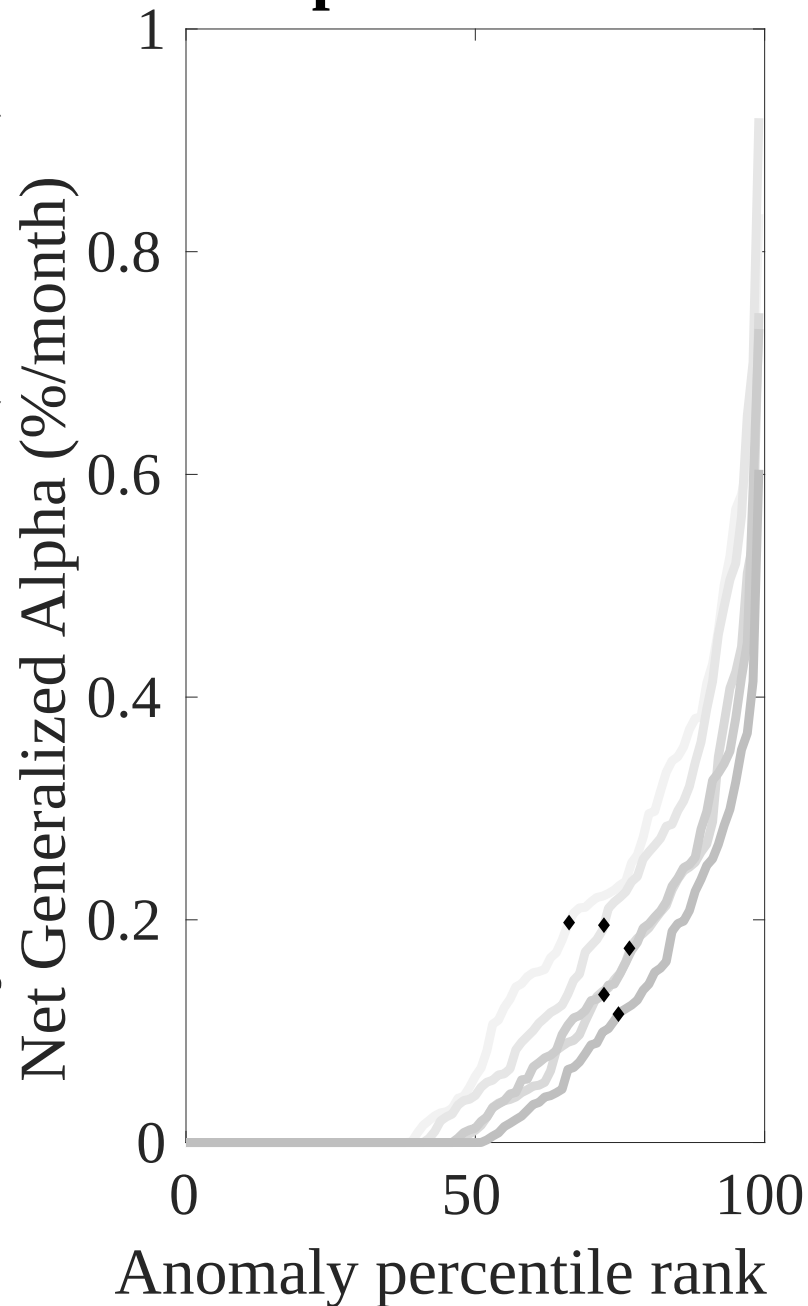
This figure plots the growth of a \$1 invested in 212 anomaly trading strategies (gray lines), and compares those with the DCM trading strategy (red line). The strategies are constructed using value-weighted quintile sorts using NYSE breakpoints. Panel A plots results for gross strategy returns. Panel B plots results for net strategy returns.

Gross Alpha Percentiles



Novy-Marx and Velikov (RFS, 2016)

Net Alpha Percentiles



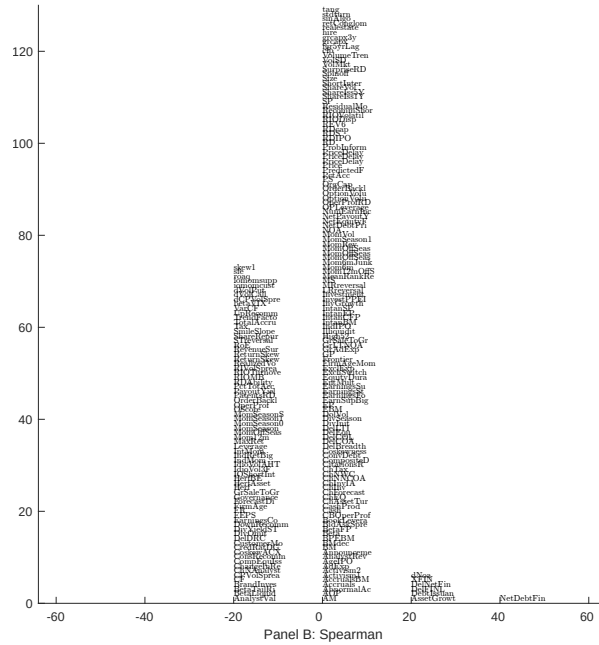
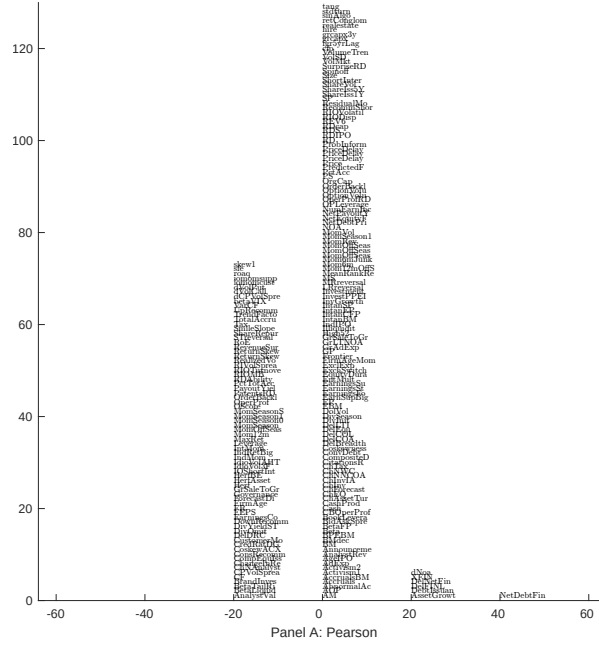


Figure 5: Distribution of correlations.

This figure plots a name histogram of correlations of 209 filtered anomaly signals with DCM. The correlations are pooled. Panel A plots Pearson correlations, while Panel B plots Spearman rank correlations.

This figure plots an agglomerative hierarchical cluster plot using Ward's minimum method and a maximum of 10 clusters.

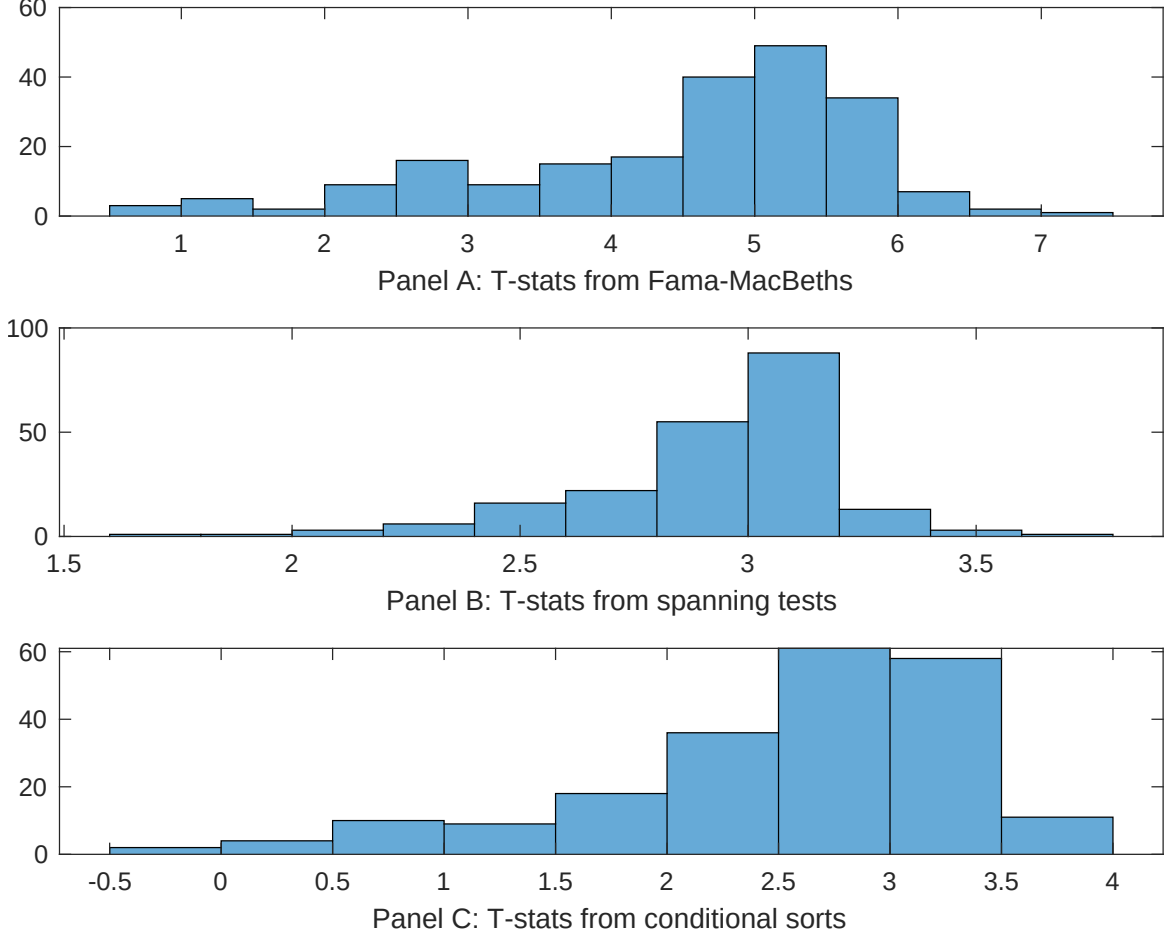


Figure 7: Distribution of t-stats on conditioning strategies

This figure plots histograms of t-statistics for predictability tests of DCM conditioning on each of the 209 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{DCM} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{DCM}DCM_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 209 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{DCM,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 209 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based on one of the 209 filtered anomaly signals at a time. Then, within each quintile, we sort stocks into quintiles based on DCM. Stocks are finally grouped into five DCM portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted DCM trading strategies conditioned on each of the 209 filtered anomalies.

Table 4: Fama-MacBeths controlling for most closely related anomalies

This table presents Fama-MacBeth results of returns on DCM. and the six most closely related anomalies. The regressions take the following form: $r_{i,t} = \alpha + \beta_{DCM}DCM_{i,t} + \sum_{k=1}^s \beta_{X_k} X_{i,t}^k + \epsilon_{i,t}$. The six most closely related anomalies, X , are Net debt financing, Change in financial liabilities, Net external financing, Inventory Growth, Asset growth, change in ppe and inv/assets. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 197306 to 202406.

Intercept	0.13 [5.35]	0.13 [5.38]	0.14 [5.72]	0.14 [5.30]	0.14 [5.79]	0.14 [5.76]	0.15 [5.74]
DCM	0.21 [2.65]	0.19 [2.78]	0.24 [3.49]	0.36 [4.28]	0.15 [1.91]	0.22 [3.01]	0.11 [1.18]
Anomaly 1	0.19 [8.72]						0.90 [1.32]
Anomaly 2		0.17 [9.20]					-0.80 [-1.66]
Anomaly 3			0.19 [6.35]				0.10 [1.80]
Anomaly 4				0.41 [6.95]			0.20 [0.34]
Anomaly 5					0.10 [8.29]		0.50 [2.33]
Anomaly 6						0.17 [7.58]	0.88 [3.14]
# months	612	612	612	612	612	612	612
$\bar{R}^2(\%)$	0	0	1	0	0	0	0

Table 5: Spanning tests controlling for most closely related anomalies

This table presents spanning tests results of regressing returns to the DCM trading strategy on trading strategies exploiting the six most closely related anomalies. The regressions take the following form: $r_t^{DCM} = \alpha + \sum_{k=1}^6 \beta_{X_k} r_t^{X_k} + \sum_{j=1}^6 \beta_{f_j} r_t^{f_j} + \epsilon_t$, where X_k indicates each of the six most-closely related anomalies and f_j indicates the six factors from the [Fama and French \(2015\)](#) five-factor model augmented with the [Carhart \(1997\)](#) momentum factor. The six most closely related anomalies, X , are Net debt financing, Change in financial liabilities, Net external financing, Inventory Growth, Asset growth, change in ppe and inv/assets. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 197306 to 202406.

Intercept	0.14 [2.09]	0.15 [2.10]	0.14 [2.03]	0.16 [2.32]	0.16 [2.26]	0.16 [2.27]	0.14 [2.01]
Anomaly 1	18.72 [4.99]						15.86 [3.08]
Anomaly 2		10.84 [2.75]					-0.78 [-0.15]
Anomaly 3			14.29 [4.13]				12.05 [3.19]
Anomaly 4				6.09 [2.25]			6.29 [2.22]
Anomaly 5					-0.72 [-0.18]		-6.87 [-1.64]
Anomaly 6						4.05 [1.29]	-1.73 [-0.52]
mkt	-5.17 [-3.28]	-4.96 [-3.10]	-3.24 [-1.96]	-5.32 [-3.32]	-5.22 [-3.24]	-5.27 [-3.28]	-3.62 [-2.18]
smb	4.42 [1.83]	4.96 [2.02]	10.66 [4.03]	6.74 [2.76]	6.13 [2.50]	6.09 [2.50]	9.80 [3.51]
hml	-10.19 [-3.35]	-10.22 [-3.30]	-9.36 [-3.03]	-11.13 [-3.60]	-10.88 [-3.50]	-11.37 [-3.64]	-8.73 [-2.82]
rmw	1.07 [0.34]	2.16 [0.68]	-5.41 [-1.45]	4.01 [1.27]	3.13 [0.99]	3.21 [1.02]	-4.89 [-1.29]
cma	20.29 [4.30]	21.51 [4.42]	15.84 [3.07]	19.73 [3.74]	25.99 [3.82]	21.92 [4.12]	17.56 [2.52]
umd	2.23 [1.41]	2.51 [1.54]	3.34 [2.12]	2.82 [1.74]	3.44 [2.15]	3.44 [2.15]	1.61 [0.99]
# months	612	612	612	612	612	612	612
$\bar{R}^2(\%)$	13	11	12	10	10	10	15

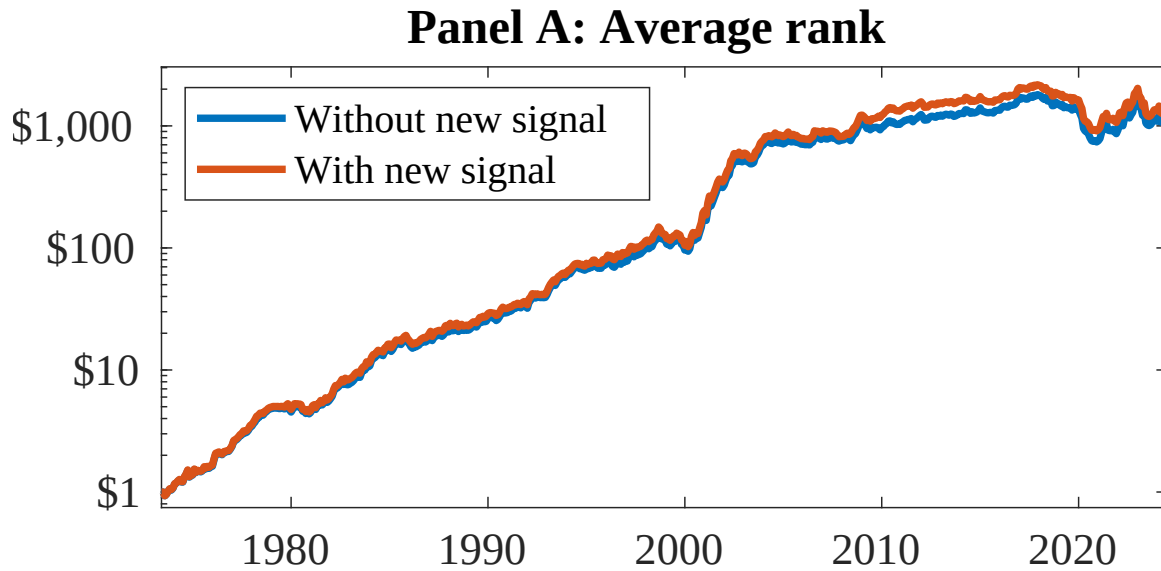


Figure 8: Combination strategy performance

This figure plots the growth of a \$1 invested in trading strategies that combine multiple anomalies following [Chen and Velikov \(2022\)](#). In all panels, the blue solid lines indicate combination trading strategies that utilize 155 anomalies. The red solid lines indicate combination trading strategies that utilize the 155 anomalies as well as DCM. Panel A shows results using "Average rank" as the combination method. See [Section 7](#) for details on the combination methods.

References

- Berk, J. B., Green, R. C., and Naik, V. (1999). Optimal investment, growth options, and security returns. *Journal of Finance*, 54(5):1553–1607.
- Carhart, M. M. (1997). On persistence in mutual fund performance. *Journal of Finance*, 52:57–82.
- Carlson, M., Fisher, A., and Giammarino, R. (2004). Corporate investment and asset price dynamics: Implications for the cross-section of returns. *Journal of Finance*, 59(6):2577–2603.
- Chen, A. and Velikov, M. (2022). Zeroing in on the expected returns of anomalies. *Journal of Financial and Quantitative Analysis*, Forthcoming.
- Chen, A. Y. and Zimmermann, T. (2022). Open source cross-sectional asset pricing. *Critical Finance Review*, 27(2):207–264.
- Cochrane, J. H. and Piazzesi, M. (2005). Bond risk premia. *American Economic Review*, 95(1):138–160.
- Detzel, A., Novy-Marx, R., and Velikov, M. (2022). Model comparison with transaction costs. *Journal of Finance*, Forthcoming.
- Fama, E. F. and French, K. R. (1993). Common risk factors in the returns on stocks and bonds. *Journal of Financial Economics*, 33(1):3–56.
- Fama, E. F. and French, K. R. (2015). A five-factor asset pricing model. *Journal of Financial Economics*, 116(1):1–22.
- Fama, E. F. and French, K. R. (2018). Choosing factors. *Journal of Financial Economics*, 128(2):234–252.

- Hou, K., Xue, C., and Zhang, L. (2015). Digesting anomalies: An investment approach. *Review of Financial Studies*, 28(3):650–705.
- Liu, L. X., Whited, T. M., and Zhang, L. (2009). Investment-based expected stock returns. *Journal of Political Economy*, 117(6):1105–1139.
- Novy-Marx, R. and Velikov, M. (2016). A taxonomy of anomalies and their trading costs. *Review of Financial Studies*, 29(1):104–147.
- Novy-Marx, R. and Velikov, M. (2023). Assaying anomalies. *Working paper*.
- Papanikolaou, D. and Kogan, L. (2011). Growth opportunities and technology shocks. *American Economic Review*, 101(3):532–536.
- Zhang, L. (2005). The value premium. *Journal of Finance*, 60(1):67–103.